Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

A. 3.0 amps, 4 ohms

B. 2.8 amps, 3.2 ohms

C. 3.75 amps, 3.2 ohms

D. 4 amps, 3 ohms

E. 3.33 amps, 3.6 ohms

F. 5 amps, 2.4 ohms

G. 4.5 amps, 2.67 ohms

H. 2.2 amps, 5.45 ohms

I. 2.5 amps, 4.8 ohms

J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>

The power dissipated by the bulb (P) is given by the equation P = V x I, where V is the voltage and I is the current. Given that the power dissipated is 40 watts and the voltage is 12 volts, we can solve for current:

I = P / V

I = 40 W / 12 V

I = 3.33 A

The resistance of the bulb can be found using the equation R = V / I:

R = V / I

R = 12 V / 3.33 A

R = 3.6 ohms

</think>

<answer>3.33 amps, 3.6 ohms</answer>